EXP 17 : EYE DIAGRAM ANALYSIS OF OPTICAL COMMUNICATION LINK USING OPTISYTEM

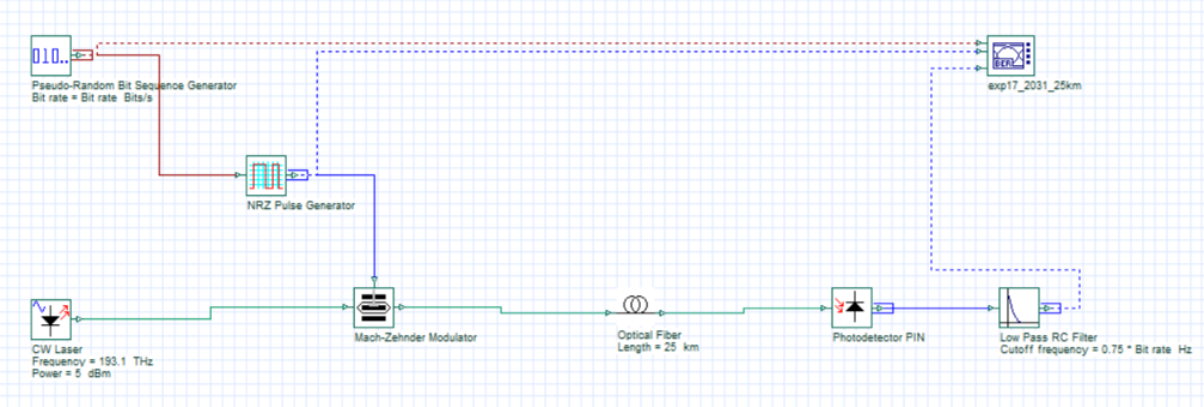
OBJECTIVES :

1. Design and establish a fiber optic communication link using OptiSystem software.
2. Measure and evaluate the Bit Error Rate (BER) of the system for different fiber lengths.
3. Study the effect of fiber length on BER, Q-factor, a enuation, and dispersion.

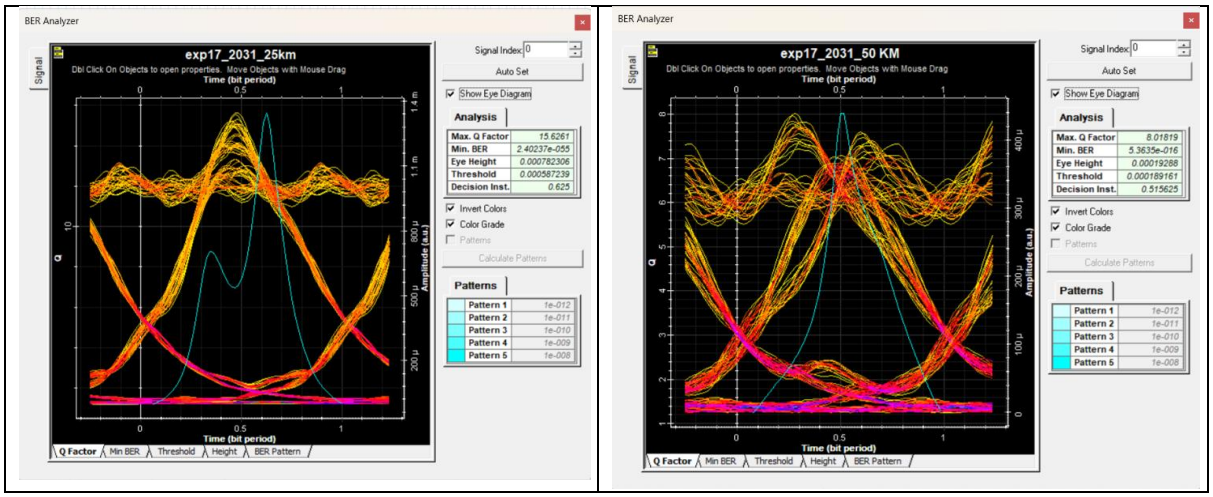
SOFTWARE REQUIRED

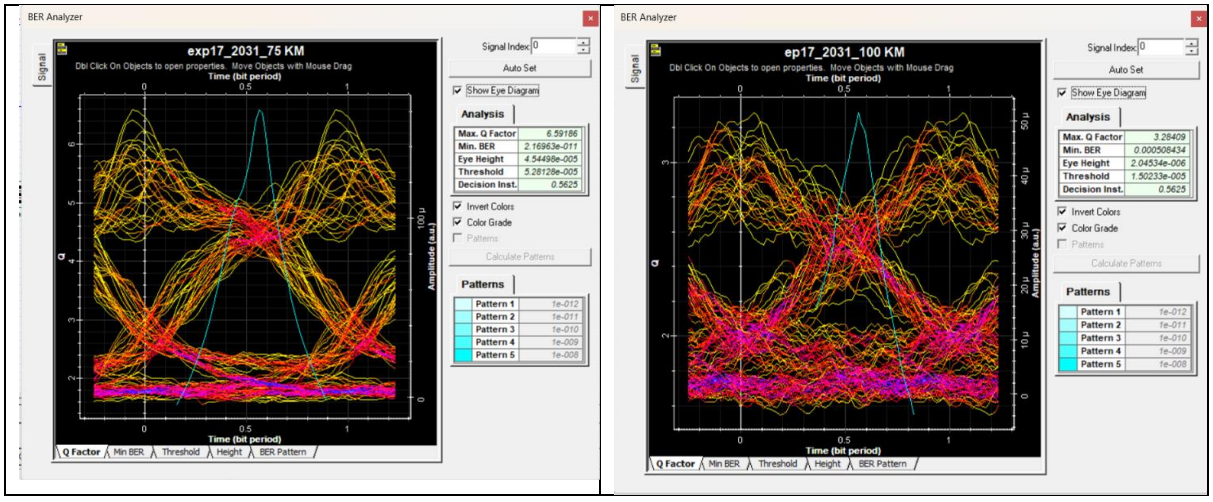
• Optisystem software

CIRCUIT DIAGRAM:



BER OUTPUTS:





RELFECTION :

I learned to design and simulate an optical fiber communication link using OptiSystem and analyze the eye diagram, Q-factor, and BER for different fiber lengths.

I observed that increasing fiber length causes signal degradation due to attenuation and dispersion, reducing Q-factor and increasing BER, which limits reliable transmission distance.

RUBRICS :

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & Fiber Link Setup	Correct selection and connection of OptiSystem components for the optical fiber link (PRBS Generator, NRZ Pulse Generator, CW Laser, Optical Fiber, PIN Photodetector, Low Pass Filter, 3R Regenerator, BER Analyzer).	All transmitter, fiber, receiver, and analyzer components are correctly selected, connected in sequence, and configured without errors.	All components are present with minor connection or layout errors, but the link remains functional.	Component selection is partially complete; minor connection or configuration errors impede signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Execution	Accurate setup of input parameters: Optical Power (5 dBm / 10 dBm), Frequency (193.1 THz), Wavelength (1550 nm), Bit Rate (10 Gbps), and Fiber Length (25 to 150 km).	All input parameters and fiber length test cases are accurately set and simulated.	Test case parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but multiple fiber length cases are missing.	Incorrect parameter settings or failure to run the required fiber length cases.	20%

Measurement & Eye Diagram Observation	Accurate extraction and recording of Max. Q-Factor, Min. BER, Eye Height, Threshold, and Decision Instant from the BER Analyzer for each fiber length.	Precise data extraction for all fiber lengths, neatly tabulated, with clear eye diagrams captured for each test case.	Data is collected with minor omissions in Q-factor, BER, or eye parameters.	Partial data collection with incomplete tables or missing eye diagrams.	Inaccurate, incomplete, or missing data and measurements .	20%
BER & Signal Degradation Analysis	Detailed evaluation of the effect of fiber length on BER, Q-factor, eye opening, attenuation, and dispersion.	Comprehensive analysis correctly relating fiber length to attenuation, dispersion, eye closure, Q-factor decline, and BER increase.	Clear analysis identifying basic trends in Q-factor, BER, and eye height versus fiber length.	Basic attempt at result analysis without explaining physical mechanisms such as attenuation and dispersion.	Erroneous or missing analysis of simulation results and eye diagrams.	20%
Report & Conclusion	Identification of the maximum reliable transmission distance, comparison across short, medium, long, and very long links, and structural clarity.	Report features clear eye diagrams, complete output tables, and insightful conclusions identifying the reliable link length limit.	Satisfactory report with relevant eye diagrams and sound concluding summary.	Incomplete report with missing diagrams, tables, or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

CONCLUSION:

The study shows how fiber length affects BER and signal quality in an optical communication system.

- 25 km fiber: highest Q-factor (15.26) and lowest BER (7.06×10^{-53}), with a wideopen eye diagram.
- 50 km fiber: reduced Q-factor (8.02) and BER of 5.26×10^{-16} due to ajenuation; the link is still reliable ($Q > 6$, $BER < 10^{-9}$).
- 75 km fiber: Q-factor falls to 4.89 and BER rises to 5.00×10^{-7} as ajenuation and dispersion close the eye; the link is no longer reliable.
- 100 km, 125 km and 150 km fiber: complete link failure (Q-factor = 0, BER = 1, eye height = 0). The maximum reliable length is therefore between 50 km and 75 km.